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Precise Measurement of the F^{19} Magnetic Moment

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Using a new circuit, a precise measurement of the frequency ratio of nuclear magnetic resonance of fluorine nucleus to that of proton in HF aq. was performed at the magnetic field of about 1800 gauss. The value $\nu_F/\nu_H=0.940816$ was obtained with the smallest error ± 0.000009 compared with the values by other research workers.

§1. Introduction

The magnetic moment of the F^{19} has been measured by many workers with the nuclear magnetic resonance method. After the tabulation by J. E. Mack,[1] new data was added by R. E. Sheriff and D. Williams.[2] The present author intended to measure the magnetic moment of various nuclei and the first result on the F^{19} is reported here. A brief report was already published in the Phys. Rev.[3]

§2. Experiments

(a) Magnetic Field.

The magnetic field is supplied by an electromagnet whose pole faces having each an area of 15 cm × 15 cm are finished to 0.01 mm and are kept parallel by three distant pieces of 40 ± 0.01 mm. length. For the modulation of the magnetic field, we use the frequency of 200 cps generated by an electromagnetic tuning fork and amplified by a push pull circuit of 6L6 tubes. To vary the D. C. magnetic field near the resonance, a sweep circuit shown in Fig. 1 is used. The range of the field varied is adjusted by a suitable choice of the tube and the sweep time is adjusted by a proper selection of the time constant (C × R) of the grid circuit. To measure the current through the magnet coil, the potential difference across the standard resistance of 0.1Ω connected series to the coil is measured by a potentiometer. Without a special stabilizer of the current, the field is stable enough under good condition of the source batteries to be able to get the right resonance figure.

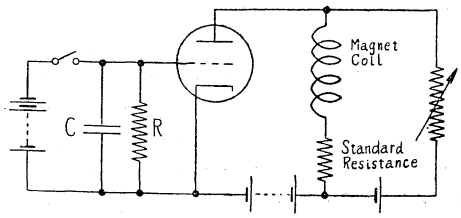


Fig. 1. The Sweep Circuit.

(b) Detector and Amplifier.

For the resonance detector, the so-called "*Inoue*'s Circuit" which had been used as a

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detector of small changes in capacitance was adopted. As shown in Fig. 2, a parallel resonance circuit ($L \times C$) is connected to the control grid of a pentode, and hf voltage of angular frequency ω generated by a triode oscillator is supplied to the screen grid. Varying the capacitance C, the plate current of the pentode suffers a sudden change near the position of $1/\sqrt{LC}=\omega$. For a nuclear resonance detector, the sample is put in the coil of this parallel resonance circuit, and so, the slight change of the inductance of the coil due to the resonance absorption gives the amplifier connected series fo the detector a corresponding voltage as its output. The amplifier is constituted by a R-C negative feedback circuit and has a resonance peak at 200 cps with a half width of 3cps. Following this amplifier is the phase sensitive detector composed of two 6SA7 tubes. The signal frequency voltage is put into the fourth grids in anti-phase and the reference voltage from the phase shifter is supplied to the first grids in-phase. The D. C. output voltage from this detector is amplified by one-stage D.C. amplifier which operates the recording meter of 10 mA.

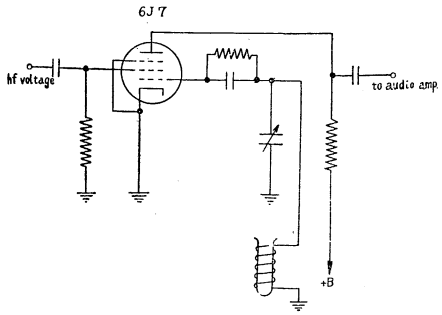


Fig. 2 The Detector Circuit.

(c) Measurement of the Frequency Ratio.

We already know, when the F^{19} resonance is operated at just 8 Mc/sec, the proton resonance appears near 8.5 Mc/sec in the same magnetic field and so these frequencies make some audio frequency basts with the harmonics of 500 kc/sec oscillator. Two similar systems each composed of oscillator, resonance detector, 200 cps amplifier, phase sensitive detector, D. C. amplifier and 10 mA recording meter are used, one for the fluorine and the other for the proton, both samples being HF aq. (20 percent solution). The hf source for the proton is a self oscillator and that for the fluorine is a crystal oscillator whose frequency is tuned to the upper 1000 cps side band of 8 Mc/sec standard wave (JJY), i e. 8.001 Mc/sec. The difference frequencies of these two oscillators to the 500 kc/sec crystal oscillator calibrated by 8 Mc/sec standard wave are measured with the bridge type precision frequency meter. The method to decide the frequency of proton resonance f_H appearing exactly at the same field with that of 8 Mc/sec fluorine is as follows: The frequency of the proton oscillator is kept slightly higher than f_H and the magnetic field is swept slowly from a higher value to a lower, so that first the proton resonance appears and next the fluorine resonance. Then the frequency of the self oscillator for the proton is quickly lowered by a few kc/sec (Δf), so the proton resonance appears again as the field becomes lower. During this process the current through the magnet coil is measured with a potentiometer and the signs are marked on the recording meter at equal intervals of current. Assuming the linearity of the field strength to the current in the region of this small sweep, f_H is decided by measuring the frequency Δf and the distances on the recording paper between the two proton resonances and the fluorine resonance.

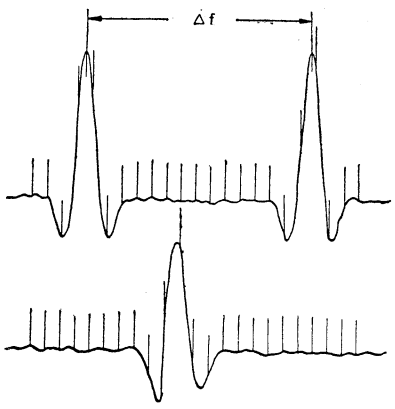


Fig. 3.

(d) Measurement of the Field Inhomogeneity.

The difference of the field strength between two samples is measured as follows: The crystal oscillator is stopped and the detector

is operated by a small voltage introduced through a coupling to the self oscillator, and thus both of two detectors are operated with the same frequency to produce the proton resonance. When the field is swept, two proton resonances appear at slightly different currents. Marking on the recorder the current at the intervals equal those in the case of the proton-fluorine resonances, and measuring the distance on the recording paper between these two proton resonances, the field inhomogeneity between the two samples is decided. It measures 0.25 gauss throughout the series.

§3. Result

Because of the same spin of the proton and the F^{19}, the frequency ratio ν_F/ν_H is equal to the magnetic moment ratio μ_F/μ_H without diamagnetic correction. Table I shows the values of μ_F/μ_H of 12 independent runs obtained by the method mentioned above. Our disclosed uncertainty is taken as three times the usual probable error. As a comparison, μ_F/μ_H values in HF sample obtained by other workers are listed together showing our uncertainty minimum.

Table I. Experimental results

Run	μ_F/μ_H	
2	0.940831	
7	0.940825	
9	0.940818	
11	0.940803	
12	0.940792	
17	0.940815	
18	0.940784	
26	0.940820	
28	0.940829	
29	0.940830	
38	0.940811	
40	0.940807	
Mean	0.940814±0.000009	
	0.940816±0.000015	Reference (4)
	0.94077 ±0.00010	(5)
	0.94075 ±0.00017	(6)
	0.94086 ±0.00005	(2)

The maximum error estimated from various errors in the experiment, for example; the uncertainty of the measurement of audio frequency and so on, amounts to the same order with the uncertainty stated.

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